LAB – 7: FUZZING FOR INPUT VALIDATION BUGS

SCREENSHOTS

Execution of the Original Buggy Application

```
PS C:\Users\Ananya V\Documents\GitHub\L6_L7_55_car_insurance_premium\Lab7> python car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 01 ab 1234
Normalized registration: KA01AB1234

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000;4500;800
Total claimed: 17300

==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999): 14/08/1999
Driver age in 2026: 27
PS C:\Users\Ananya V\Documents\GitHub\L6_L7_55_car_insurance_premium\Lab7>
```

Identification of Input Validation Bugs Using Hypothesis

Command: python -m pytest test_car_insurance_premium_parser.py -v

```
=========================== short test summary info ============================
FAILED test_car_insurance_premium_parser.py::test_normalize_registration_no_crash - AssertionError: normalize_registration crashed: 'NoneType' object has no attribute 'strip'
FAILED test_car_insurance_premium_parser.py::test_total_claim_amount_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_car_insurance_premium_parser.py::test_driver_age_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================== 3 failed in 8.21s ===============================
```

Execution of the Fixed Application

```
PS C:\Users\Ananya V\Documents\GitHub\L6_L7_55_car_insurance_premium\Lab7> python car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 01 ab 1234
Normalized registration: KA01AB1234

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000;4500;800
Total claimed: 17300

==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999): 14/08/1999
Driver age in 2026: 27
```

Successful Execution of Hypothesis Tests on the Fixed Application

```
PS C:\Users\Ananya V\Documents\GitHub\L6_L7_55_car_insurance_premium\Lab7> python -m pytest test_car_insurance_premium_parser.py -v
================================== test session starts ==================================
platform win32 -- Python 3.13.1, pytest-8.3.4, pluggy-1.5.0 -- C:\Program Files\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Ananya V\Documents\GitHub\L6_L7_55_car_insurance_premium\Lab7
plugins: anyio-4.10.0, hypothesis-6.170.14, cov-7.1.0
collected 3 items

test_car_insurance_premium_parser.py::test_normalize_registration_no_crash PASSED      [ 33%]
test_car_insurance_premium_parser.py::test_total_claim_amount_no_crash PASSED          [ 66%]
test_car_insurance_premium_parser.py::test_driver_age_no_crash PASSED                  [100%]

=================================== 3 passed in 1.43s ===================================
```